McKell Woodland

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Education

Rice University, Houston, TX — Aug 2024
Ph.D. Computer Science

Brigham Young University, Provo, UT — Apr 2018
B.S. Applied and Computational Mathematics

Professional Experience

The University of Texas MD Anderson Cancer Center, Houston, TX — Jan 2020 – Present
Data Scientist; Graduate Research Assistant

- Researching out-of-distribution detection and generative modeling evaluation in medical imaging
- Peer-reviewed publications: MELBA, MICCAI 2024, UNSURE 2023, Diagnostics, SASHIMI 2022, & Eur. J. Radiol.
- Awards: Outstanding Reviewer – Honorable Mention (MICCAI 2024), Best Spotlight Paper (UNSURE 2023), Graduate Student Travel Grant, Young Investigator Award – Significance (PBDW 2021), and Worldwide Innovations in Medical Physics Scholarship (WIMP 2022)
- Invited talks: RSNA 2025, TMI Symposium, BYU Computer Science, MSB Thesis Madness (MICCAI 2024), UNSURE 2023, AAPM 2021-2023, & MedPhys Slam (SWAAPM 2022)
- Participated in a public debate at PBDW 2023; Spoke on a MICCAI Student Board panel
- Moderated session at 2nd Annual Research Retreat in Oncological Data and Computational Sciences

Microsoft, Redmond, WA (remote) — May 2020 - Aug 2020
Data Science Intern

- Devised a similarity score in Python to analyze overlap between suspicious behavior detectors
- Produced, using Kusto, a tool for Microsoft Threat Experts to use the score to detect malicious behavior
- Moderated a Q&A session with the TF and CTO of Microsoft's AI Cognitive Services

Lawrence Livermore National Laboratory, Livermore, CA — May 2019 - Aug 2019
Data Science Research and Development Intern

- Developed an event detection model for videos by embedding an R2U-Net into a ConvLSTM using PyTorch
- Crafted 100,000 colorful moving MNIST with obscurations videos with OpenCV (VRAM)
- Gave final model (99% acc on VRAM) to client to apply to classified data with obscured & moving objects

Perception Control and Cognition Lab, Provo, UT — Jan 2018 – May 2019
Graduate Research Assistant

- Trained a Rainbow DQN on game states embedded by a CVLAE to reduce sample complexity in RL
- Awards: Session Winner (BYU SRC 2019); Invited Talk: BYU SRC 2019

3M Health Information Systems, Murray, UT — May 2018 – Aug 2018
Data Science Intern

- Made an RNN on an EC2 instance to predict over 50,000 diagnoses from 1.7 million documents using Keras
- Built the basis of a deep learning framework in Keras to predict DRG codes
- Assembled a framework to analyze the text behind any amount of medical notes

National Security Agency, Laurel, MD — May 2017 – Aug 2017
Director's Summer Program Intern

- Implemented custom convolutional neural nets using Python and received TS/SCI clearance
- Analyzed an inertial navigation problem with noisy data and submitted an internal technical paper
- Prepared a briefing for the Deputy Director and presented findings to the IDA at Princeton

Brigham Young University, Provo, UT — Jan 2016– Dec 2017
Research Assistant; Developer

- Researched optimal facility location, developable conical mechanisms, and instructional technology
- Peer-reviewed publications: J. Mech. Robot, JSSR, & IRRODL; Invited talks: IDETC/CIE 2020, NERCCS 2019, & BYU SRC 2018
- Utilized Python, Mathematica, & SQL technologies

Other Research Experience

Texas Children's Hospital; Medical Informatics Corp, Houston, TX — Aug 2019 – Dec 2019

- Created a novel classification algorithm to detect the onset of a heart rhythm disorder with 90% AUROC
- Engineered a novel waveform visualization tool (using t-SNE) for use in a development pipeline
- Team took first place in Rice's Data2Knowledge Showcase; poster presentation in WFPICCS 2020

Computer Science Department, BYU, Provo, UT — Aug 2017– Dec 2017

- Extracted 23 features from web-scraped BYU speeches and used a wrapper algorithm for feature selection
- Ran 4 machine learning models on the features and achieved 79% accuracy on the speeches' popularity
- Invited talks: NCUR 2018

Peer-Reviewed Articles

- **Woodland, M.**, Patel, N., Castelo, A., Al Taie, M., Eltaher, M., Yung, J.P., Netherton, T.J., Calderone, T.L., Sanchez, J.I., Cleere, D.W., Elsaiey, A., Gupta, N., Victor, D., Beretta, L., Patel, A.B., & Brock, K.K. (2024). Dimensionality Reduction and Nearest Neighbors for Improving Out-of-Distribution Detection in Medical Image Segmentation. *Machine Learning for Biomedical Imaging*, 2(UNSURE2023 special issue), 2006-2052. doi.org/10.59275/j.melba.2024-g93a
- Baroudi, H. Brock, K.K., Cao, W., Chen, X., Chung, C., Court, L.E., El Basha, M.D., Farhat, M., Gay, S., Gronberg, M.P., Gupta, A.C., Hernandez, S., Huang, K., Jaffray, D.A., Lim, R., Marquez, B., Nealon, K., Netherton, T.J., Nguyen, C.M., Reber, B., Rhee, D.J., Salazar, R.M., Shanker, M.D., Sjogreen, C., **Woodland, M.**, Yang, J., Yu, C., & Zhao, Y. (2023). Automated Contouring and Planning in Radiation Therapy: What Is 'Clinically Acceptable'? *Diagnostics*, 13, 667. doi.org/10.3390/diagnostics13040667
- Sen, A., Troncoso, P., Venkatesan, A., Pagel, M.D., Nijkamp, J.A., He, Y., Lesage, A., **Woodland, M.**, & Brock, K.K. (2021). Correlation of In-Vivo Imaging with Histopathology: A Review. *European Journal of Radiology*, 144, 109964. doi.org/10.1016/j.ejrad.2021.109964
- **Woodland, M**., Hsiung, M., Matheson, E., Safsten, A., Greenwood, J., Halverson, D.M., & Howell, L.L. (2021). Analysis of the Rigid Motion of a Developable Conical Mechanism. *Journal of Mechanisms and Robotics*, 13, 031106-1 to 031106-8. doi.org/10.1115/1.4050294
- Kimmons, R., Hunsaker, E., Jones, E.J., & **Stauffer, M.** (2019). The nationwide landscape of K-12 school websites in the United States: Systems, services, intended audiences, and adoption patterns. *International Review of Research in Open and Distributed Learning*, 20(3). doi.org/10.19173/irrodl.v20i4.3794
- Kimmons, R., McGuire, K., **Stauffer, M.**, Jones, E.J., Gregson, M., & Austin, M. (2017). Religious identity, expression, and civility in social media: Results of data mining Latter-day Saint Twitter accounts. *Journal for the Scientific Study of Religion*, 56(3), 180-210. doi.org/10.1111/jssr.12358

Conference Proceedings

- **Woodland, M.,** Castelo, A., Al Taie, M., Albuquerque Marques Silva, J., Eltaher, M., Mohn, F., Kundu, S., Yung, J.P., Patel, A.B., & Brock, K.K. (2024). Feature Extraction for Generative Medical Imaging Evaluation: New Evidence Against an Evolving Trend. *Medical Image Computing and Computer Assisted Intervention 2024.*
- **Woodland, M.**, Patel, N., Al Taie, M., Yung, J.P., Netherton, T.J., Patel, A.B., & Brock, K.K. (2023). Dimensionality Reduction for Improving Out-of-Distribution Detection in Medical Image Segmentation. In: Sudre, C.H., Baumgartner, C.F., Dalca, A., Mehta, R., Qin, C., Wells, W.M. (eds) *Uncertainty for Safe Utilization of Machine Learning in Medical Imaging 2023*. LNCS, vol 14291. Springer, Cham. doi.org/10.1007/978-3-031-44336-7_15
- **Woodland, M.**, Wood, J., Anderson, B.M., Kundu, S., Lin, E., Koay, E., Odisio, B., Chung, C., Kang, H.C., Venkatesan, A.M., Yedururi, S., De, B., Lin, Y.-M., Patel, A.B., & Brock, K.K. (2022). Evaluating the Performance of StyleGAN2-ADA on Medical Images. In: Zhao, C., Svoboda, D., Wolterink, J.M., Escobar, M. (eds) *Simulation and Synthesis in Medical Imaging 2022.* LNCS, vol 13570. Springer, Cham. doi.org/10.1007/978-3-031-16980-9_14
- **Woodland, M.**, Hsiung, M., Matheson, E.L., Safsten, C.A., Greenwood, J., Halverson, D.M., & Howell, L.L. (2020). Analysis of the Rigid Motion of a Developable Conical Mechanism. *International Design Engineering Technical Conferences & Computers and Information in Engineering 2020*. doi.org/10.1115/DETC2020-22643

- Henrichsen, A., Jafek, B., O'Bryant, J., & **Stauffer, M.** (2018). Brigham Young University Speeches Popularity Predictor. *National Conference on Undergraduate Research 2018*. libjournals.unca.edu/ncur

Under Review

- **Woodland, M.,** Al Taie, M., O'Connor, C., Lebimoyo, O., Yung, J.P., Kinahan, P., Fuller, C.D., Odisio, B.C., Patel, A.B., & Brock, K.K. (2025). Generative Modeling for Interpretable Failure Detection in Liver CT Segmentation and Scalable Data Curation of Chest Radiographs.

Abstracts

- **Woodland, M.,** O'Connor, C., Wood, J., Patel, A.B., & Brock, K.K. (2023). Interpretable Out-of-Distribution Detection with Generative Adversarial Networks. *Medical Physics*, 50(6), e154-155. doi.org/10.1002/mp.16525
- **Woodland, M.**, Wood, J., O'Connor, C., Patel, A.B., & Brock, K.K. (2022). StyleGAN2-based Out-of-Distribution Detection for Medical Imaging. In: *Medical Imaging meets NeurIPS 2022.* sites.google.com/view/med-neurips-2022/abstracts
- Reber, B., van Dijk, L.V., Anderson, B.M., Mohamed, A.S., Rigaud, B., He, Y., **Woodland, M.,** Fuller, C.D., Lai, S.Y., and Brock, K.K. (2022) Comparison of Machine Learning and Deep Learning Methods for the Prediction of Osteoradionecrosis Resulting from Head and Neck Cancer Radiation Therapy. *International Journal of Radiation Oncology Biology Physics*, 114(3), e124. doi.org/10.1016/j.ijrobp.2022.07.946
- **Woodland, M.,** Wood, J., Anderson, B.M., Kundu, S., Lin, E., Koay, E., Odisio, B., Chung, C., Kang, H.C., Venkatesan, A., Yedururi, S., De, B., Lin, Y., Patel, A.B., & Brock, K.K. (2022). Comparing Transfer Learning, Data Augmentation, and Data Expansion in the Improvement of Medical Image Generation. *Medical Physics*, 49(6), 134. doi.org/10.1002/mp.15769
- **Woodland, M.,** Patel, A.B., Anderson, B.M., Lin, E., Koay, E., Odisio, B., & Brock, K.K. (2021). GAN-Driven Anomaly Detection for Active Learning in Medical Imaging Segmentation. *Medical Physics*, 48(6). doi.org/10.1002/mp.15041
- Reber, B., Anderson, B.M., Mohamed, A., Van Dijk, L., Rigaud, B., McCulloch, M., He, Y., **Woodland, M**., Fuller, C., Lai, S., & Brock, K.K. (2021). Predicting Osteoradionecrosis from Head and Neck Radiotherapy Using a Residual Convolutional Neural Network. *Medical Physics*, 48(6). doi.org/10.1002/mp.15041
- Sailsbery, M., Heiner, J., Robertson, C., **Stauffer, M.**, & Jarvis, T. (2019). Facility Location Using Markov Chains on Spatial Networks. In: *Northeast Regional Conference on Complex Systems 2019*. coco.binghamton.edu/nerccs

Invited Talks

- *Radiological Society of North America 2025.*
- Predicting Segmentation Model Failure with Out-of-Distribution Detection and Generative Adversarial Networks. *Tumor Measurement Initiative Symposium 2025.*
- Predicting Liver Segmentation Model Failure with Feature-based Out-of-Distribution Detection and Generative Adversarial Networks. *Computer Science Colloquia at Brigham Young University, 2024.*
- Predicting Liver Segmentation Model Failure with Feature-Based Out-of-Distribution Detection and Generative Adversarial Networks. *MICCAI Student Board PhD Thesis Madness 2024.*
- Dimensionality Reduction for Improving Out-of-Distribution Detection in Medical Image Segmentation. *Uncertainty for Safe Utilization of Machine Learning in Medical Imaging 2023.*
- Interpretable Out-of-Distribution Detection with Generative Adversarial Networks. *American Association of Physicists in Medicine 2023.*
- Comparing Transfer Learning, Data Augmentation, and Data Expansion in the Improvement of Medical Image Generation. *American Association of Physicists in Medicine 2022.*
- Houston, Our AI Models Have a Problem! *Southwest Regional Chapter of the American Association of Physicists in Medicine 2022.*
- GAN-Driven Anomaly Detection for Active Learning in Medical Imaging Segmentation. *American Association of Physicists in Medicine 2021.*
- Understanding LDS Twitter use. *Instructional Design and Learning Community Conference 2016*.

Posters

- Feature Extraction for Generative Medical Imaging Evaluation: New Evidence Against an Evolving Trend. *Medical Imaging Computing and Computer Assisted Intervention 2024.*
- Dimensionality Reduction for Improving Out-of-Distribution Detection in Medical Image Segmentation. *Uncertainty for Safe Utilization of Machine Learning in Medical Imaging 2023.*
- Interpretable OOD Detection using GANs: An Application to the MIDRC Data Repository. *Practical Big Data Workshop 2022*.
- StyleGAN2-based Out-of-Distribution Detection for Medical Imaging. *Medical Imaging meets NeurIPS 2022.*
- Evaluating the Performance of StyleGAN2-ADA on Medical Images. *Simulation and Synthesis in Medical Imaging 2022.*
- StyleGAN2-based Out-of-Distribution Detection in Computed Tomography. *Practical Big Data Workshop 2022*.
- Improving the Generation of Synthetic Medical Images using Data Augmentation and Transfer Learning. *Southwest Regional Chapter of the American Association of Physicists in Medicine 2022.*
- Detecting Out-of-Distribution Images for Active Learning using a Generative Adversarial Network (StyleGAN2). *Practical Big Data Workshop 2021.*
- Comparison of Machine Learning Methods for the Prediction of Osteoradionecrosis (ORN) from Head and Neck Cancer (HNC) Radiation Therapy. *Practical Big Data Workshop 2021.*
- GAN-Driven Anomaly Detection for Active Learning in Medical Imaging Segmentation. *Winter Institute of Medical Physics 2021.*
- A Novel Algorithm for Early Detection of Junctional Ectopic Tachycardia in Patients with Congenital Heart Disease*. World Federation of Pediatric Intensive & Critical Care Societies 2020*.

Panel Participant

- Panel Discussion on the trips and tips of MICCAI Review, Rebuttal, Challenge, and Workshop. *MICCAI Student Board Webinar Series, 2024*.
- "And You're Out!" – Setting Minimum Expectations for Big Data/AI Abstracts*. Practical Big Data Workshop 2023*.

Awards

(2024) Outstanding Reviewer – Honorable Mention. *Medical Image Computing and Computer Assisted Intervention 2024.*
(2023) Best Spotlight Paper. *Uncertainty for the Safe Utilization of Machine Learning in Medical Imaging 2023.*
(2023) Graduate Student Travel Grant. *Rice Engineering Alumni.*
(2021) Early Career Investigator Significance Award. *Practical Big Data Workshop 2021.*
(2021) Worldwide Innovations in Medical Physics Finalist. *Winter Institute of Medical Physics 2021.*
(2019) Showcase Winner. *Rice Data2Knowledge Showcase.*
(2019) Session Winner. *Brigham Young University Student Research Conference.*
(2019) Y-Serve Exceptional Leader. *Brigham Young University Y-Serve.*
(2018) Outstanding Mathematician. *Brigham Young University Math Department.*
(2017) Outstanding Mathematician. *Brigham Young University Math Department*.

Technical Skills

Languages: Python, C++, Java, HTML, and C
Platforms: Docker, Kubernetes, Linux, GitHub, Amazon Web Services, Spark, SQL, Kusto, and LaTeX
Python Packages: PyTorch, TensorFlow, PySpark, Scikit-Learn, Keras, OpenCV, Autograd, and Pandas

Teaching Experience

The Coding School, Remote Aug 2019 – Present
Lead Instructor; Developer; Young Professional Advisory Board Member
- Taught introductory Python and AI courses to over 400 elementary, middle, and high school students
- Instructed 300+ HBCU and community college professors on how to introduce AI into their classrooms
- Tutored 5 students one-on-one in data structures, OOP, algorithms, and AI in both Java and Python
- Authored the deep learning and SQL curriculum and edited the machine learning curriculum
- Aided in the design of the governance structure of the Young Professional Advisory Board

Computer Science, Brigham Young University - Idaho, Remote Apr 2022 – July 2022

Online Adjunct Instructor

- Taught the online pilot course CSE 450: Machine Learning and Data Mining
- Served as the Online Course Representative for CSE 450
- Received 12% higher reviews than all other online instructors in terms of concern, timeliness, & enthusiasm

Computer Science, Rice University, Houston, TX

Teaching Assistant; Instructional Coach Oct 2020 – Apr 2022

- TA'd: Statistics for Computing and Data Science (Fall 2021), Introduction to Computer Security (Spring 2021), and Programming for Data Science (Fall 2020)
- Help students prepare to give a research presentation in the department's Graduate Research Seminar

Mathematics, Brigham Young University, Provo, UT Jan 2017 – Dec 2017

Curriculum Developer

- Wrote 7 Python labs for the Applied & Computational Mathematics curriculum: iterative solvers, profiling, SymPy, SQL 1 & 2, one-dimensional optimization, & gradient descent
- Worked with collaborators to edit over 15 different labs

Academic Services

Reviewer, Applied Sciences, 2025
Reviewer, Medical Image Computing & Computer Assisted Intervention, 2024-2025
Reviewer, Communications Medicine, 2024
Reviewer, Uncertainty for Safe Utilization of Machine Learning in Medical Imaging, 2024
Reviewer, Workshop on Deep Generative Models for Health held in conjunction with NeurIPS, 2023

Community Involvement

Rice University, Houston, TX Aug 2019 – Aug 2024

Panelist, Mentor, Teaching Assistant

- Spoke on a graduate student panel for the Rice CS/IO club
- Mentored undergraduate students in the 2022 Rice Datathon
- Tutored international online master's students who were struggling with the data science curriculum

Summer STEM Institute, Remote Jun 2021 – Aug 2021

Research Mentor

- Advised a high-school student who built a glioblastoma detector on MRI scans using a pre-trained CNN
- Guided another student who built an arrhythmia detector on ECG signals using a pre-trained ResNet
- Delivered a talk on deep learning in medical imaging to ~300 high school students

BYU Y-Serve: Kids Who Code, Provo, UT Aug 2018 - Apr 2019

Executive Director

- Established a program that facilitates university students teaching kids in the local community coding skills
- Facilitated a Girls Who Code club (20+ girls) at an elementary school (taught Scratch)
- Received a top leadership award from the Y-Serve organization

Utah STEM Action Center, Salt Lake City, UT Apr 2018 – May 2019

Ambassador

- Promoted STEM at 10+ events and called 100+ schools to teach them about STEM opportunities
- Spoke on a panel at a "Girls Who Code" event at Adobe to 200+ middle school girls
- Accepted certificate from Utah Governor Gary Herbert for service

BYU Women in Computer Science, Provo, UT Jan 2018 – Dec 2018

Public Outreach Officer

- Planned a hackathon to help women at BYU have projects for interviews
- Interacted with companies to promote opportunities among the female computer science students
- Increased the number of social media followers by 500%